

Vector Floors: Farkas-Certified Obstruction under Multiple Sustainability Constraints

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

September 23, 2026

Abstract

Sustainability constraints are rarely single floors: minimum stocks, maximum emissions, service levels, and institutional limits bind simultaneously. The obstruction calculus to date states its certificates for one constraint at a time. This paper, the D4 item of the programme’s generalization catalogue, states the common-action obstruction for *vector* floors $\mathcal{V} = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$ and shows that its static polyhedral form carries an exact dual certificate: the stacked safe-action system is infeasible precisely when a Farkas multiplier vector exists, so the obstruction itself is a checkable object of rational arithmetic. On the programme’s two-patch working system the dynamic vector-floor reading reproduces the audited structure verdicts (blind and aggregate readings nonviable; z_1 -only, z_2 -only, and full readings viable), adding a floor strictly shrinks the viable pair family ($28 \rightarrow 36$ reversed: one floor admits 36, two floors 28), and a three-floor stacked instance exhibits the complementarity reading of the multipliers (inactive row, multiplier zero). All claims are machine-verified in exact rational arithmetic; the verification script ships with the paper.

Introduction and relation to the programme

The programme’s completed items — the obstruction-calculus manuscript with its four certificate families, the continuous-to-finite companion bridge, the viable-selector unification, the stochastic selector, and the five-layer successor absorbing general information structures and exit-time values — share one working simplification: the safe set is presented through a single scalar margin. The source catalogue for the present generalization notes that this is the least sustainable reading of the theory: real constraint systems are vectors of floors. Its proposed generalization is adopted here verbatim as the object of study: the vector safe set, the vector common-action correspondence, and the polyhedral special case in which the obstruction becomes a finite linear infeasibility with a Farkas certificate. Nothing in the present paper modifies the completed items; it re-reads their working systems under vector floors and certifies the new static layer exactly.

The vector-floor model

Let the state space carry m floors $q_1, \dots, q_m$ with safe set

$$\mathcal{V} = \{x : q_j(x) \geq 0 \text{ for every } j\}.$$

For a state x with instrument set $U(x)$, the *vector safe-action set* collects the instruments that keep every floor non-negative under every admissible disturbance:

$$\mathcal{R}_{\mathcal{V}}(x) = \{\nu \in U(x) : x' \in \mathcal{V} \text{ for all successors } x'\}.$$

For a belief B (the set of states compatible with the reading), the common-action condition with per-cell semantics reads $\mathcal{R}_{\mathcal{V}}^B(B) = \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x)$ within each indistinguishability cell, exactly as in the successor’s read-then-act convention. The vector obstruction is the static statement

$$\mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies B \notin \mathcal{K}_{\mathcal{I}},$$

where $\mathcal{K}_{\mathcal{I}}$ is the observation-based viability kernel under information structure $\mathcal{I}$: a single moment without a jointly floor-preserving instrument certifies nonviability, no matter how capable the future.

Proposition 1 (Branchwise/joint gap for vector floors). If for each state $x \in B$ separately $\mathcal{R}_{\mathcal{V}}(x) \neq \emptyset$ but $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$, then every branch of B is individually floor-servable while the belief is vector-obstructed: the information constraint, not any single floor, is the cause of nonviability.

Proof. The branchwise hypothesis provides, for each x , an instrument ν_x keeping x in $\mathcal{V}$; hence no floor fails in isolation. The joint hypothesis denies the existence of one instrument admissible for all x simultaneously, which is precisely the per-cell read-then-act requirement; any observation-based policy must move the whole cell with one instrument and therefore fails at the first step. The cause cannot be a floor: each is satisfiable alone. It cannot be the future: the failure occurs in one step. It is the pairing of one instrument with several states, i.e. the information constraint. $\square$

The Farkas certificate

Let the floors and the instrument set be polyhedral, so that each safe-action set is $\mathcal{R}_{\mathcal{V}}(x_i) = \{u : A_i u \leq b_i\}$ with rational data. The common safe-action system stacks to $Au \leq b$, and:

Proposition 2 (Obstruction as dual infeasibility). The stacked system $Au \leq b$ is infeasible if and only if there exists a rational $\lambda \geq 0$ with $\lambda^\top A = 0$ and $\lambda^\top b < 0$ (Farkas’ lemma). Consequently the vector obstruction admits a certificate that is itself checkable in exact rational arithmetic: given λ , the three identities are verifiable by Fraction arithmetic with no solver and no search.

Proof. Farkas’ lemma gives the equivalence. Checkability: the certificate side requires evaluating two matrix products and one inequality — exact operations on rationals. The infeasibility side is certified, not searched: $\lambda^\top A = 0$ and $\lambda^\top b < 0$ with $\lambda \geq 0$ yield $0 = \lambda^\top (Au) \leq \lambda^\top b < 0$ for every candidate u , a contradiction. $\square$

Remark 1 (Complementarity reading). In the stacked system of a three-floor instance below, the certified multiplier assigns weight 0 to a redundant row: inactive constraints receive zero multipliers, the standard complementary pairing of primal activity and dual weight, here read as “the obstruction is carried by exactly the floors that bind.”

Verified instances

All of the following are re-derived and checked by the verification script (6/6 exact check families; standard library, Fraction arithmetic, deterministic).

E1 (stacked equivalence, both directions). The two-floor instrument instance $\mathcal{R}_1 = \{u : u \leq \frac{2}{5}\}$, $\mathcal{R}_2 = \{u : u \geq \frac{3}{5}\}$ stacks to rows $u \leq \frac{2}{5}$, $-u \leq -\frac{3}{5}$; the multiplier $\lambda = (1, 1)$ satisfies $\lambda^\top A = 0$ and $\lambda^\top b = \frac{2}{5} - \frac{3}{5} = -\frac{1}{5} < 0$, certifying infeasibility; a 101-point rational grid confirms no u satisfies both. The feasible contrast $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{1}{5}\}$ (with a redundant third row $u \leq \frac{4}{5}$) admits no certificate — the exact case analysis forces $\lambda_1 = \lambda_2$ and then $\lambda^\top b \geq 0$ — and the witness $u = \frac{1}{5}$ is exhibited. Both directions of Proposition 2 are thereby exercised.

E2 (the belief-level gap). On the belief merging a state with floor-set $\{u \leq \frac{2}{5}\}$ with a state with floor-set $\{u \geq \frac{3}{5}\}$: branchwise witnesses $u = \frac{2}{10}$ and $u = \frac{6}{10}$ exist; the joint set is empty with the E1 certificate. Proposition 1 fires exactly.

E3 (dynamic vector floors). On the programme’s two-patch system the safe set is already a vector object $q_1 : z_1 \geq 1$, $q_2 : z_2 \geq 1$ ($m = 2$) served by one shared instrument. Per-cell read-then-act viability over the audited beliefs $\{(1, 2), (2, 1)\}$ and $\{(1, 2), (2, 1), (2, 2)\}$ reproduces the audited verdicts under five information structures: blind nonviable, aggregate nonviable, z_1 -only viable, z_2 -only viable, full viable — the successor’s structure table re-read as a statement about two floors and one instrument.

E4 (floor monotonicity). Dropping the second floor (q_1 only) enlarges the viable pair family from 28 to 36 of 36 under full information, setwise, with strict separation witnessed (e.g. $\{(1, 1), (1, 2)\}$: viable on one floor, nonviable on two). Adding floors destroys viability — the marginal cost of each additional institutional constraint is a computable set.

E5 (exact multiplier recheck). Every multiplier used above is re-validated componentwise ($\lambda \geq 0$; $\lambda^\top A = 0$; $\lambda^\top b < 0$) in Fraction arithmetic.

E6 (three floors, complementarity). The stacked system $\{u \leq \frac{2}{5}\} \wedge \{u \geq \frac{3}{5}\} \wedge \{u \leq \frac{9}{10}\}$ is infeasible with certificate $\lambda = (1, 1, 0)$: the inactive third row carries multiplier zero, exhibiting the complementarity reading of Remark 1.

Pareto readings of trade-offs between floors. The Farkas layer is static; the companion bridge’s continuous-to-finite transfer and the successor’s timing layer are cited, not re-derived. Non-convex instrument codices invalidate Proposition 2 (Farkas’ lemma is a convexity statement); the monitoring-design companion of this catalogue records the exact non-convex counterinstance. Sustainability floors with uncertain thresholds are treated in the programme’s buffered-viability companion, not here.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `vector_floor_certificates_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.

What is not claimed

The dynamic vector-floor theory is not developed beyond the per-cell re-reading of the audited systems: no vector-valued comparison functions, no multi-floor dynamic programming, no